Impact on emerging technologies: IoT, 5G, and edge computing

Wi-Fi 6 and 7 are well-suited for supporting emerging technologies like the Internet of Things (IoT), edge computing, and complementing 5G networks.

- ***IoT applications:*** In smart homes and cities, where the number of connected devices is increasing, the network's ability to manage these connections efficiently is critical. Wi-Fi 6's power-saving features and range improvements make it a transformative technology for IoT, allowing devices like security cameras, smart thermostats, and appliances to stay connected over extended periods with minimal energy consumption.
- ***Edge computing:*** As edge AI applications expand, particularly in real-time video processing, low latency becomes essential. Wi-Fi 7's latency reduction and MLO will enable faster decision-making at the edge, allowing applications such as automated sorting in agriculture or real-time object recognition in industrial settings to perform more efficiently.
- ***Smart cities:*** With a higher number of IoT devices deployed across cities, from traffic lights to environmental sensors, Wi-Fi 6 provides the necessary network efficiency. Meanwhile, Wi-Fi 7's low-latency and higher throughput capabilities can handle data-heavy tasks, such as real-time analytics for transportation systems.

Balancing power efficiency with performance: Wi-Fi 6 and 7 aim for superior performance but face power efficiency challenges due to growing IoT devices. Features like OFDMA and MU-MIMO increase power demands. To address this, energy-efficient microcontrollers and power management ICs use dynamic voltage scaling, intelligent idle-state management, and target wake time (TWT) to optimise power use.

Antenna design for higher frequencies: The 6GHz spectrum in Wi-Fi 6E and 7 allows high-bandwidth communication but complicates antenna design. Efficient multi-band antennas (2.4GHz, 5GHz, and 6GHz) must support wide channels (up to 320MHz in Wi-Fi 7) while minimising interference. Advanced antenna modules and RF front-end solutions ensure optimal performance, using techniques like antenna diversity, beamforming, and modular designs for easier integration.

Reducing latency for real-time applications: Wi-Fi 7's multi-link operation (MLO) offers ultra-low latency, which is ideal for AR/VR, gaming, and industrial automation. However, maintaining low latency with high throughput in congested environments is challenging. Network solutions prioritise real-time traffic with advanced QoS features, smart scheduling, and optimised channel access to reduce delays and ensure efficient performance.

Addressing thermal management challenges: With Wi-Fi 6 and 7, increased data rates and processing demands generate significant heat in compact devices. Effective thermal management is critical to prevent performance loss or hardware failure. Infineon's power management solutions include thermal protection systems, efficient power distribution, and materials with high thermal conductivity to ensure reliable operation and prevent overheating.

Looking ahead: Connectivity and microcontrollers

The future of connected devices, especially in the IoT space, will increasingly rely on advanced microcontrollers that integrate both connectivity and computing capabilities. Infineon and other industry leaders are already building microcontrollers optimised for low-power, high-range IoT applications. These new connected microcontrollers will allow devices to operate reliably even at the farthest points of the network, ensuring consistent performance regardless of placement.

Wi-Fi 6 and Wi-Fi 7 are not just about faster data speeds; they represent a fundamental shift in how networks manage multiple devices, handle latency, and conserve energy. With innovations like MU-MIMO, OFDMA, and MLO, these technologies are poised to support the future of smart homes, IoT, and edge computing, ensuring a connected world that is faster, more reliable, and more efficient. The continued evolution of wireless technology promises to make real-time, low-latency applications a reality, paving the way for new innovations and seamless connectivity in every aspect of our lives.

The advancements in Wi-Fi 6 and the upcoming Wi-Fi 7 are transforming wireless communication to meet the needs of modern users. From handling crowded networks to enabling next-gen real-time applications, these technologies are paving the way for a faster, more efficient, and responsive wireless future. As devices continue to proliferate, both Wi-Fi 6 and Wi-Fi 7 will be key enablers in the journey towards seamless, ubiquitous connectivity.

Wi-Fi 6 and Wi-Fi 7 are set to revolutionise the wireless landscape with unprecedented performance and capabilities. However, these benefits come with a host of design challenges, from managing RF complexities to balancing power efficiency and handling heat dissipation. Infineon offers a range of solutions that address these challenges, enabling manufacturers to deliver innovative products that fully capitalise on the advantages of these next-generation wireless technologies.

By understanding these challenges and adopting the right technologies and design principles, engineers can successfully create products that meet the demanding requirements of Wi-Fi 6 and Wi-Fi 7. END

This article is based on an interaction between Shivaram, vice president of Wi-Fi Products, Infineon Technologies, and Akanksha Sondhi Gaur, senior technology journalist at EFY. It was transcribed and curated by Akanksha.

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